

ML evaluation pipeline

End-to-end evaluation workflow · 4 phases · 9 stages · Dagster orchestrated

Data preparation Evaluation setup Analysis & metrics QA & reporting

PHASE 1 — DATA PREPARATION & LABELING

1

Annotation

Multi-source data labeling using internal tooling and automated OCR extraction across all annotation types

internal-tools OCR

ANNOTATION TYPES — SUB-FLOWS

Internal annotation tools OCR text extraction Bounding box NER tagging Image segmentation Classification labels Key-value extraction

2

Dataset export

Orchestrated export pipeline writing annotated datasets to BigQuery via Dagster jobs with schema validation

BigQuery Dagster

EXPORT PIPELINE



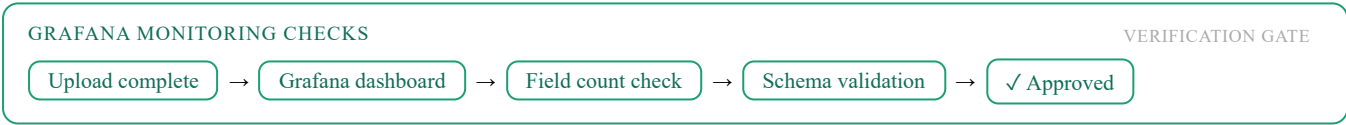
3

Cloud storage upload

Dataset uploaded to GCS bucket; completeness and field integrity verified via Grafana monitoring dashboard

GCS Grafana

GRAFANA VERIFICATION SUB-FLOW



PHASE 2 — EVALUATION SETUP

4

Evaluation config setup

Define evaluation parameters, scoring thresholds, metric targets, and baseline configurations for each run

YAML config-store

CONFIG COMPONENTS

Model parameters Score thresholds Metric definitions Baseline targets Dataset split config Run environment

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Dagster evaluation workflow

End-to-end orchestration of the evaluation job with dependency management, scheduling, and live run monitoring

Dagster cloud-run

PIPELINE EXECUTION STAGES



PHASE 3 — ANALYSIS & METRICS

6

Prediction comparison

Systematic comparison of model predictions against ground truth annotations at scale across evaluation splits

BigQuery pandas

COMPARISON PIPELINE



7

Metrics generation

sklearn custom-eval

Compute classification, detection, and extraction metrics across all evaluation splits and annotation types

COMPUTED METRICS

Accuracy Precision Recall F1 score AUC-ROC Confusion matrix mAP (detection) IoU score OCR accuracy

PHASE 4 — QA & REPORTING

8

QA validation

review-tool checklist

Human and automated review gate with threshold-based sign-off before report generation proceeds

VALIDATION FLOW

Auto checks -> Peer review -> Threshold gate -> Lead sign-off -> ✓ Approved

9

Final evaluation report

Grafana PDF BigQuery

Consolidated report with full metrics, findings, and visual dashboards distributed to all stakeholders

DELIVERABLES

PDF report Grafana dashboard BigQuery results table Stakeholder share Archive & version tag